

Security Features in Real-World Applications

1. Identify five real-world applications or online services involving communication, authentication, or transactions. For each application, identify the security feature, mechanism, protocol/technology, and explain how it provides security.

Answer

- Today, almost every online service need security because users share personal information, communicate with others, or make financial transactions. Different applications use different security features depending on what they need to protect.
- The following are five common real-world applications and the security technologies they use.

Application	Security Feature	Security Mechanism	Protocol / Technology	How it provides security
Gmail	Authentication	Password + Multi-Factor Authentication	OAuth 2.0 / OpenID Connect, HTTPS/TLS	Verifies the user's identity and protects login and account communication.
WhatsApp	Confidentiality	End-to-End Encryption	Signal Protocol	Encrypts messages so that only the sender and intended recipient can read them.
Online Banking	Confidentiality	Encryption	HTTPS / TLS	Encrypts information such as login details and transaction data while it travels between the user and bank server.
Google Pay	Authentication and Confidentiality	Device authentication and encryption	HTTPS/TLS, tokenization	Protects payment information and helps ensure that only the

				authorized user can make a payment.
Amazon	Authentication and Integrity	Account authentication and secure communication	HTTPS/TLS, MFA	Protects account information and ensures that information exchanged between the user and the service is not easily modified or exposed.

a) Gmail

- Gmail contains important personal information, so authentication and confidentiality are very important.
- A user normally signs in using a password, and additional verification such as multi-factor authentication can provide another layer of protection. Secure authentication technologies such as OAuth 2.0 and OpenID Connect are also used in Google's identity ecosystem.
- HTTPS/TLS protects communication between the user's device and Google's servers.
- In simple words, even if someone knows the user's email address, additional authentication makes it harder for that person to access the account.

b) WhatsApp

- WhatsApp is mainly a communication application, so protecting the contents of messages is very important.
- WhatsApp uses end-to-end encryption based on the Signal Protocol. The message is encrypted on the sender's device and is decrypted on the recipient's device.
- This means that the message is not normally readable by intermediaries while it is being transmitted.

c) Online Banking

- Online banking handles highly sensitive information such as account details, passwords and financial transactions.

- Banks use HTTPS with TLS to protect communication between the customer's device and the bank's server. Authentication mechanisms such as passwords, PINs, OTPs and multi-factor authentication may also be used.
- TLS encrypts the data while it is travelling through the network.

d) Google Pay

- Digital payment applications need strong authentication and protection of payment information.
- Google Pay uses security technologies including encryption, device-based authentication and tokenization in supported payment scenarios. HTTPS/TLS is used to protect communication with online services.
- Tokenization can replace sensitive payment information with a token, reducing the need to expose the actual payment details during a transaction.
- Instead of repeatedly exposing sensitive payment information, security mechanisms help protect the actual payment credentials and verify the user before a payment is made.

e) Amazon

- Amazon handles user accounts, addresses, orders and payments, so both authentication and data protection are important.
- Amazon uses HTTPS/TLS to protect communication between the user's browser or application and its services. Account security can also include passwords and multi-factor authentication.
- TLS helps protect data from being easily read or modified while travelling between the user's device and the server.
- When a user logs in, places an order or accesses account information, secure communication helps protect the information being exchanged.

2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Answer

- Security controls are measures used by applications and organizations to protect their systems and users from attacks.

- The three basic security goals considered here are:
 - Confidentiality: Prevent unauthorized people from accessing information.
 - Integrity: Prevent or detect unauthorized modification of information.
 - Availability: Make sure that services and information are available when users need them.
- Security controls can also be understood based on what they do:
 - Prevent: Stop an attack or reduce its chance of succeeding.
 - Detect: Identify suspicious activity or an attack.
 - Recover: Restore services or data after an attack or failure.

The following five applications are good examples:

Application	CIA Property	Security Control	Mechanism / Protocol	Prevent, Detect or Recover	How it Helps
Gmail	Confidentiality	Encryption	HTTPS/TLS	Prevent	Protects email and account data during transmission.
Gmail	Confidentiality	MFA	Authentication systems / OTP / security keys	Prevent	Makes unauthorized account access more difficult.
WhatsApp	Confidentiality	End-to-end encryption	Signal Protocol	Prevent	Protects message contents from unauthorized access during communication.
Online Banking	Confidentiality & Integrity	Secure communication	HTTPS/TLS	Prevent	Protects banking information while it travels between the user and bank.

Online Banking	Integrity	Transaction verification	OTP / MFA / transaction authentication	Prevent	Helps ensure that important transactions are authorized by the customer.
Google Pay	Confidentiality	Tokenization	Payment tokenization	Prevent	Reduces exposure of sensitive payment credentials during transactions.
Google Pay	Availability	Fraud and risk monitoring	Automated monitoring systems	Detect	Suspicious transactions can be identified and blocked or reviewed.
Amazon	Confidentiality	Account authentication	Password + MFA	Prevent	Reduces the possibility of unauthorized access to accounts.
Amazon	Availability	Backup and recovery systems	Redundancy / backup	Recover	Helps restore services or information after failures or attacks.

a) Gmail

Gmail needs to protect emails, attachments and account information.

- Confidentiality:
HTTPS/TLS protects information while it is being transmitted. Authentication and MFA help prevent unauthorized people from accessing an account.
- Prevention:
MFA makes it harder for an attacker to enter an account even if the password has been compromised.

- **Detection:**
Suspicious login activity can be detected by account security systems.
- **Recovery:**
Account recovery mechanisms allow legitimate users to regain access if they lose access to their account.

b) WhatsApp

WhatsApp mainly needs to protect communication between users.

- **Confidentiality:**
End-to-end encryption based on the Signal Protocol protects message contents.
- **Integrity:**
Cryptographic mechanisms used by the protocol help ensure that messages cannot simply be modified without detection.
- **Prevention:**
Encryption prevents an attacker who intercepts network traffic from simply reading the message contents.
- **Detection:**
Authentication and security notifications can help users identify changes related to account or device security.
- **Recovery:**
Users can re-register their accounts and use available account recovery/security features if their account is compromised.

c) Online Banking

Online banking is one of the strongest examples of the need for security because it deals directly with money.

- **Confidentiality:**
HTTPS/TLS protects login credentials and other sensitive information during communication.
- **Integrity:**
Transaction authentication mechanisms help ensure that important transactions are authorized by the customer.
- **Prevention:**
MFA, OTPs, transaction verification and account protections make unauthorized transactions more difficult.

- **Detection:**
Banks monitor transactions for unusual patterns, such as unexpected locations, devices or transaction behaviour.
- **Recovery:**
Banks can block accounts or cards, investigate fraudulent transactions and restore access after an incident, depending on the situation.

d) Google Pay

Payment applications need to protect both the user's account and payment information.

- **Confidentiality:**
Encryption and secure communication protect information from unauthorized access.
- **Integrity:**
Authentication and transaction controls help ensure that payments are made by the authorized user.
- **Prevention:**
Device authentication, screen locks, payment authentication and tokenization reduce the risk of unauthorized payments.
- **Detection:**
Payment systems can monitor transactions and identify unusual or suspicious activity.
- **Recovery:**
If suspicious activity is detected, transactions or payment methods may be blocked or investigated, depending on the circumstance.

e) Amazon

Amazon stores information such as user accounts, addresses, orders and payment-related information.

- **Confidentiality:**
HTTPS/TLS protects information exchanged between the user's device and Amazon's services.
- **Integrity:**
Authentication and secure communication help protect account and order information from unauthorized changes.
- **Prevention:**
Passwords, MFA and other account security controls reduce unauthorized account access.

- **Detection:**
Unusual account activity and suspicious transactions can be monitored by security systems.
- **Recovery:**
Backup systems, account recovery processes and service redundancy help restore information or services after failures or security incidents.

Prevent, Detect and Recover:

Security controls can be understood more easily with a simple example.

Suppose an attacker tries to access an online banking account.

- **Prevent:**
The bank uses passwords, MFA and secure communication to make the attack more difficult.
- **Detect:**
The bank's monitoring systems identify unusual login or transaction behaviour.
- **Recover:**
The bank can block the affected account, investigate the incident and help the customer restore secure access.

This shows that good security does not only mean preventing attacks. A system should also be able to detect attacks and recover from them.